

1) Why is exploration more important in Mountain Car than in Pendulum?

Mountain Car's reward is sparse and the goal requires a *counter-intuitive* strategy (back up the left hill to build momentum) that's unreachable without substantial exploration; many naïve policies get stuck in the valley. Pendulum receives dense feedback every step, so gradient-like improvements arise with much less exploration pressure. Hence MC benefits from a more aggressive ϵ schedule.

2) Why does Pendulum require a more fine-grained discretization than Mountain Car?

Pendulum control needs smooth torques and accurate angle/velocity resolution to keep the pole near upright; coarse bins or few discrete torques induce chattering and poor stabilization. Mountain Car tolerates coarser bins because its objective is to escape the valley rather than finely regulate a near-equilibrium. Thus Pendulum uses finer state and action grids.